

TABLE I
FIXED CAPITAL

LAND AND BUILDING

Description	Details
Built-up area	500sqm
Office and stores	150sqm
Factory shed	200sqm
Assembling and testing	1500sqm
Rent payable (₹ per annum)	120,000

MACHINERY AND EQUIPMENT

Description	Indian/Imported	Quantity	Rate (₹)	Value (₹)
Bench drilling machine (½)	Indian	1	15,000	15,000
Digital multimeter (3½ digits)	Indian	3	9,000	27,000
Oscilloscope (0MHz -20MHz)	Indian	1	50,000	50,000
IC tester/EPROM programmer	Indian	1	50,000	50,000
Digital LCR meter	Indian	1	45,000	45,000
Load cell simulator	Imported	1	90,000	90,000
Portable grinder	Indian	1	20,000	20,000
Power supply (0-30V, 2A)	Indian	2	10,000	20,000
Standard weights (brass)	Indian	LS	LS	30,000
Standard weights (brass)	Indian	1	4,500	4,500
UV eraser	Indian	1	6,000	6,000
Variac (4A)	Indian	2	10,000	20,000
Total cost				377,500
OTHER FIXED ASSETS:				
Electrification charge @ 10% on machinery and equipment				37,750
Office furniture and equipment				100,000
Tools, dies, jigs and fixtures				60,000
Pre-operative expenses				30,000
Total fixed capital				605,250

Energy conservation

With growing energy needs and shortage coupled with rising costs, a greater thrust in energy efficiency in the industrial sector has been given by the government of India since 1980s. Energy Conservation Act, 2001 was enacted on August 18, 2001. It provides for efficient use of energy, its conservation and capacity building.

Following steps may help with the conservation of electrical energy:

1. Adoption of energy-conserving technologies, production aids and testing facilities
2. Efficient management of process/manufacturing machineries and systems, QC and testing equipment for yielding maximum energy conservation
3. Optimum use of electrical energy for heating during soldering process could be obtained by using efficient temperature-controlled soldering and

de-soldering stations

4. Periodical maintenance of motors, compressors, etc

5. Use of power factor correction capacitors. Proper selection and layout of lighting systems, timely switching on/off lights, use of compact fluorescent lamps wherever possible and so on

Raw materials and equipment

An electronic weighing system comprises basic load cell, suitable signal conditioners and output recorders/indicators, since both give analogue and digital outputs for further processing.

Other materials required are cabinet/housing (metal), capacitors, fluorescent displays, ICs, mechanical hardware, noise filters, rectifiers, PCBs, resistors, transformers, transistors, wires, cable connectors, consumables and packing material, among others.

As a manufacturer, procure some

basic machinery and equipment for this business. This includes:

- Bench drilling machine
- Digital multimeter
- Oscilloscope (0MHz - 20MHz)
- IC tester/EPROM programmer
- Digital LCR meter
- Load cell simulator
- Portable grinder
- Standard weights (brass)
- Multimeter (analogue)
- UV eraser

In addition, arrange utilities like water and electricity. Further, a medium- and large-scale unit demands 2000sqm area for operation.

Business economics

An indicative analysis has been given on the following basis and presumptions:

1. Basis for calculation of production capacity has been taken on a single shift of eight hours per day basis on 75 per cent efficiency.

2. Maximum capacity utilisation on single-shift basis for 300 days a year has been considered. During the first and second years of operation, capacity utilisation is 60 and 80 per cent, respectively. The unit is expected to achieve full capacity utilisation from third year onwards.

3. Salaries and wages, cost of raw materials, utilities, civil construction, etc are based on the prevailing rates in and around north India. These cost factors are likely to vary with time and location.

4. Interest on term loan and working capital loan has been taken at the rate of 16 per cent on average. This rate may vary depending upon the policy of financial institutions/agencies from time to time.

5. Cost of machinery and equipment refer to a particular make or model. Prices are approximate.

6. Break-even point percentage indicated is of full capacity utilisation.

7. Project preparation cost, whenever required, can be considered under pre-operative expenses.

8. Essential production machinery and test equipment required for the project have been indicated. The unit may also utilise common test facilities available at Electronics Test and Development Centres (ETDCs) and Electronic Regional Test Laboratories (ERTLs) set up by state governments and STQC,